

Exception

- program is running on and something goes wrong and try to catch the error or exception and inform or display error message which can understand by user is difficult.
- in python use try/except construct which provide solution

```
try:
```

```
    f = open('test.txt')
```

```
    s = f.read()
```

```
    f.close()
```

```
except IOError:
```

```
    print("Cannot open the file")
```

- exception use to identify runtime error
- during file access
- during accessing index, list when outside bound of the list
- Exception is near the top of the tree and will catch almost any exception.
- Except section can also have separate sections for catching different types of exception and handling each in different way.
- if exception is not specify any exception class, all exceptions will be caught.

```
list= [1,2,3]
try:
list[8]
exception:
    print("out of range")
else:
    print("in range")
finally:
    print ("always do this")
```

the else clause will be run if there is no exception, and the finally clause will be run whether there is an exception or not.

- whenever an exception occurs , get more information about it using the exception object, which is available only if you use the as keyword.

```
list= [1,2,3]
```

```
try:
```

```
list[8]
```

```
except Exception as e:
```

```
    print("out of range")
```

```
    print(e)
```

```
...
```

```
out of range
```

```
list index out of range
```